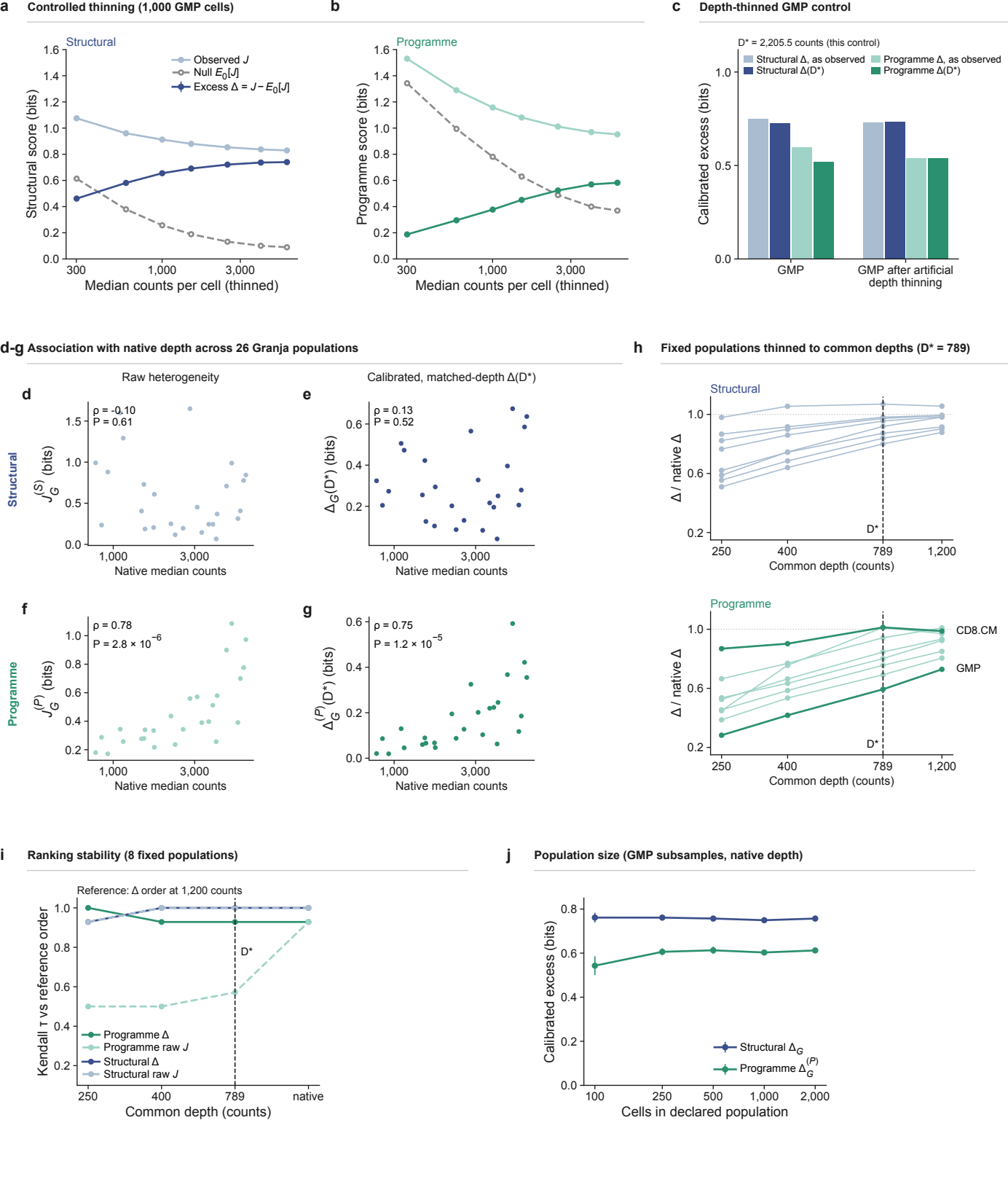

a
Controlled thinning (1,000 GMP cells)
Structural
Observed J
Null E0[J]
Excess Δ = J − E0[J]
Structural score (bits)
Median counts per cell (thinned)
b
Programme
Programme score (bits)
Median counts per cell (thinned)
c
Depth-thinned GMP control
D* = 2,205.5 counts (this control)
Structural Δ, as observed
Structural Δ(D*)
Programme Δ, as observed
Programme Δ(D*)
Calibrated excess (bits)
GMP
GMP after artificial depth thinning
d-g
Association with native depth across 26 Granja populations
Raw heterogeneity
Calibrated, matched-depth Δ(D*)
d
ρ = -0.10
P = 0.61
Native median counts
e
ρ = 0.13
P = 0.52
f
ρ = 0.78
P = 2.8 × 10−6
g
ρ = 0.75
P = 1.2 × 10−5
h
Fixed populations thinned to common depths (D* = 789)
Δ / native Δ
Common depth (counts)
CD8.CM
GMP
i
Ranking stability (8 fixed populations)
Reference: Δ order at 1,200 counts
Kendall τ vs reference order
Programme Δ
Programme raw J
Structural Δ
Structural raw J
native
j
Population size (GMP subsamples, native depth)
Structural ΔG
Programme ΔG(P)
Cells in declared population